Physics-Informed Multi-Quantile Forecasting for Risk-Averse NHS Critical-Care Surge Allocation

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Abstract. NHS critical-care planning during epidemic waves must pre-position surge capacity and route patients across regions before demand is known. We propose a forecast-to-decision pipeline that couples a per-region physics-informed forecaster on the SEI_aI_sHCRD system to a risk-averse, scenario-weighted linear program (LP) for bed-surge allocation. The forecaster uses an equally weighted multi-quantile pinball loss, including a cost-relevant $q^{(0.9)}$ branch that penalises under-prediction nine-to-one, together with a level-and-trend anchor. On a seven-region NHS England case study (Aug 2020 to Aug 2022), before point recalibration the forecaster cuts 14-day root-mean-square error (RMSE) by 19% over per-origin refitted autoregressive integrated moving average (ARIMA) and by 37% over a neural gated recurrent unit (GRU). Raw 80% interval coverage declines from 94% at 7 days to 64% at 28 days, so the three quantiles drive the LP as explicit stress scenarios rather than calibrated probability bins. With unconstrained transfers all budget-exhausting policies tie on coverage and the risk-averse formulation wins on transfer burden; once transfers are constrained the proportional heuristics concede up to 60% more scenario-weighted unmet demand.

Keywords: physics-informed neural networks, quantile forecasting, pinball loss, probabilistic forecasting, risk-averse optimisation, linear programming, NHS England

1 Introduction

NHS critical-care surge planning during epidemic waves balances the cost of prepositioning capacity against the risk of unmet demand while keeping inter-regional transfers manageable across the seven NHS England regions. COVID-19 placed sustained pressure on adult mechanical-ventilation capacity, making this a coupled forecasting and allocation problem under regime shift. Existing forecasting work spans statistical and recurrent baselines [2], graph-based epidemic and hospital-demand models [8,7], and physics-informed neural networks (PINNs) [14,1,13]; interval and quantile evaluation is also standard in epidemic forecast hubs [4]. The closest intensive-care-unit (ICU) occupancy comparator, a bidirectional long

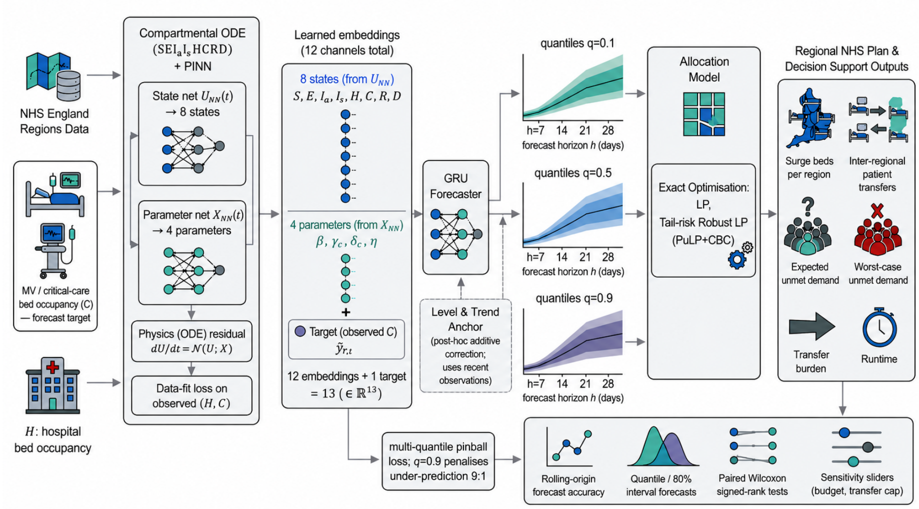

Fig. 1. Pipeline for NHS critical-care surge planning, from regional data and physics-informed forecasting to scenario-based risk-averse allocation and evaluation.

short-term memory (LSTM) network on Lazio data [15], like most such work, targets a single health system rather than multi-region planning. These forecasters are usually optimised for symmetric statistical accuracy, although critical-care planning has an asymmetric error cost. Under-prediction risks stranded patients, whereas over-prediction wastes prepared capacity.

On the allocation side, robust facility-location and health-resource models handle demand uncertainty with mixed-integer linear programming (MILP), two-stage optimisation, and metaheuristics [6,16]. Bertsimas et al. [3] couple DELPHI dynamics to a mixed-integer program for vaccination-site location, while Luo and Stellato [10] embed neural-ODE dynamics in a facility-location model. Our setting differs from Smart Predict-then-Optimize [5]. The forecaster is not trained through allocation regret, but its multi-quantile pinball objective supplies a cost-relevant upper quantile to a risk-averse optimisation layer. To our knowledge, no prior work combines physics-informed forecasting of NHS regional critical-care demand with multi-quantile training and an exact risk-averse linear-programming allocation across the seven NHS England regions.

This paper presents a forecast-to-decision pipeline (Figure 1). A per-region PINN on the SEI_aI_sHCRD ordinary-differential-equation (ODE) system is pre-trained, and its twelve outputs (eight compartmental states, four transmission and clinical parameters) are frozen and passed to a temporal GRU head that emits the horizon-wise quantiles $q^{(0.1)}$, $q^{(0.5)}$, $q^{(0.9)}$ under an equally weighted multi-quantile pinball loss; its $q^{(0.9)}$ branch penalises under-prediction nine-to-one, with a level-and-trend anchor. These quantiles are discretised into three scenarios with weights $\pi = (0.2, 0.6, 0.2)$, which deterministic and risk-averse linear programs solve exactly via PuLP against three heuristic baselines on a shared LP-slave oracle. We then report operational metrics and a sensitivity analysis over the forecaster, budget B, tail weight λ_3, travel cap τ, and transfer cap κ, together with direct calibration and sharpness metrics for the raw quantile scenarios.

We make two main contributions. (1) **Multi-quantile forecaster.** A multi-quantile pinball loss and a level-and-trend anchor cut h=14 RMSE by 19% over a refitted ARIMA model and 37% over a per-region GRU, the architecture's clearest edge; a model-agnostic per-origin recalibration then extends competitiveness to h=7–21 over the unrecalibrated baselines (level with refitted ARIMA at h=28, where seasonal-naive leads). Direct quantile evaluation distinguishes those point-forecast results from the unrecalibrated scenarios consumed by the LP. (2) **Risk-averse LP allocation.** With unconstrained transfers, the hybrid-regional risk-averse LP ties every budget-exhausting policy on coverage and wins on transfer burden; constraining transfers (a sparser network or a capped transfer service) breaks the tie, leaving the proportional rules to concede up to 60% more scenario-weighted unmet demand while the LP is never beaten on coverage.

2 Forecasting Module

2.1 SEIRD Compartmental Model

For each region $r \in \mathcal{R}$ with population N_r we model the local epidemic with an eight-compartment SEI_aI_sHCRD system whose states are the susceptible S_r, exposed E_r, asymptomatic and symptomatic infectious $I_{a,r}$ and $I_{s,r}$, hospitalised H_r, critical-care C_r, recovered R_r, and deceased D_r populations of region r, each a count of people and together summing to N_r. Figure 2 shows the flow topology, in which the distinct probabilities ω ($I_s \to H$) and ϕ ($H \to C$) disambiguate the two clinical thresholds. The system evolves as

$$\begin{aligned}
\dot{S}_r &= -\beta_r S_r (I_{s,r} + I_{a,r})/N_r + \eta_r R_r, \quad \dot{E}_r = \beta_r S_r (I_{s,r} + I_{a,r})/N_r - \alpha E_r, \\
\dot{I}_{s,r} &= \alpha \rho E_r - d_s I_{s,r}, \quad \dot{I}_{a,r} = \alpha(1-\rho)E_r - d_a I_{a,r}, \\
\dot{H}_r &= d_s \omega I_{s,r} - (d_H + \mu) H_r, \quad \dot{C}_r = \phi\, d_H H_r - (\gamma_{c,r} + \delta_{c,r}) C_r, \\
\dot{R}_r &= d_s(1-\omega) I_{s,r} + d_a I_{a,r} + (1-\phi) d_H H_r + \gamma_{c,r} C_r - \eta_r R_r, \\
\dot{D}_r &= \mu H_r + \delta_{c,r} C_r,
\end{aligned} \tag{1}$$

subject to the conservation constraint $S_r + E_r + I_{a,r} + I_{s,r} + H_r + C_r + R_r + D_r = N_r$, where a dot denotes a time derivative ($\dot{S}_r = \mathrm{d}S_r/\mathrm{d}t$). The fixed clinical constants $(\alpha, \rho, d_s, d_a, d_H, \omega, \mu, \phi)$ take the values in [1, Table 1]; rates have units of day^{-1} and probabilities lie in $[0, 1]$. Data determine the time-varying, region-specific transmission, critical-care recovery, death, and immunity-waning rates $(\beta_r, \gamma_{c,r}, \delta_{c,r}, \eta_r)$.

2.2 Per-Region PINN-SEIRD

The PINN consists of two multilayer perceptrons (MLPs) following [1], a state network U^r_{NN} (five hidden layers of 20 units) and a parameter network X^r_{NN} for

$(\beta_r, \gamma_{c,r}, \delta_{c,r}, \eta_r)$ (three hidden layers of 20 units), with tanh hidden activations and sigmoid outputs. The loss combines a data fit on the two observable compartments with an ODE residual against Equation (1). The observed compartments are H (daily general-and-acute hospital bed occupancy) and C (mechanical-ventilation [MV], i.e. critical-care, bed occupancy), the only states with daily regional NHS counts; C is the quantity the temporal head forecasts and the allocation consumes as demand ξ.

$$\begin{aligned}\mathcal{L}_r^{PINN} = &\sum_{t\in\mathcal{T}_{\text{train}}} \sum_{k\in\{H,C\}} \left(U^r_{NN,k}(t) - y_{r,t,k}/N_r\right)^2 \\ &+ \lambda_{ode} \sum_{t\in\mathcal{T}_{\text{coll}}} \left\| \frac{dU^r_{NN}(t)}{dt} - \mathcal{N}(U^r_{NN}; X^r_{NN}) \right\|^2,\end{aligned} \tag{2}$$

where $\mathcal{N}$ is the right-hand side of Equation (1), derivatives use PyTorch autograd at 256 collocation points sampled uniformly in $\mathcal{T}_{\text{coll}}$, and observations are population-normalised. We pre-train each region's PINN for 1,500 Adam steps with $\lambda_{ode} = 0.1$, $\text{lr} = 5\times10^{-3}$; both terms converge to the 2–8 $\times\, 10^{-6}$ range across regions.

With only H and C observed, the four learnable parameters $(\beta_r, \gamma_{c,r}, \delta_{c,r}, \eta_r)$ are not jointly identifiable ($\gamma_{c,r}$ and $\delta_{c,r}$ appear additively in $\dot{C}_r$), so we make no per-trajectory interpretability claim. The four-channel output is consumed as a learned feature vector by the GRU; the "w/o PINN params" ablation confirms it carries signal (Section 5.1; h=14 RMSE 14.46→27.25). Structural identifiability under $\{H, C\}$ observation is future work.

2.3 Temporal Head with Level-and-Trend Anchor

Once pre-trained the PINN is frozen and evaluated, never refit, at every origin. The downstream GRU and anchor absorb extrapolation beyond its Alpha/Delta fit. For each lookback window of length $L = 28$ ending at origin t, the augmented feature sequence is $\mathbf{f}_{r,i} = [U^r_{NN}(i) \,\|\, X^r_{NN}(i) \,\|\, \tilde{y}_{r,i}] \in \mathbb{R}^{13}$, $i \in \{t-L+1, \ldots, t\}$. Here

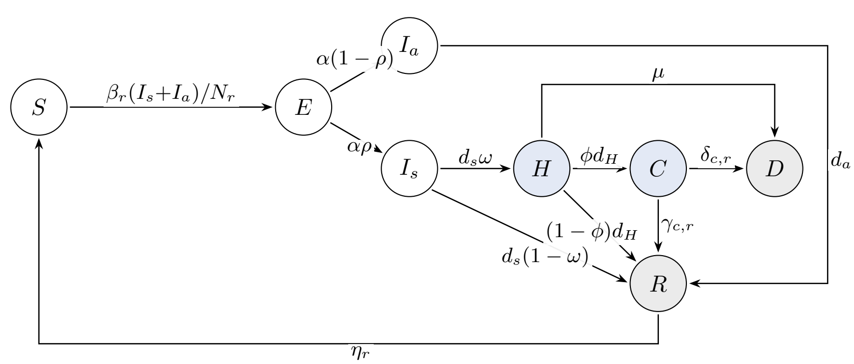

Fig. 2. SEI_aI_sHCRD compartmental model used by the physics-informed forecaster.

$U^r_{NN}(i)$ contributes all eight compartments (including the never-observed latent S, E, I_a, I_s, R, D), so the head receives the full physics-inferred state, and $\tilde{y}_{r,i}$ is the target z-scored with training-period statistics only. A two-layer GRU (hidden 128, dropout 0.15) decodes the final hidden state via a per-horizon two-layer MLP (width 64, Gaussian-error-linear-unit [GELU] activations) into standardised quantile deltas $\Delta^{h,q}_{r,t}$ for $h \in \mathcal{H} = \{7, 14, 21, 28\}$ and $q \in \mathcal{Q} = \{0.1, 0.5, 0.9\}$.

A level-and-trend anchor makes the GRU's job residual:

$$\hat{\zeta}^{h,q}_{r,t} \;=\; \Delta^{h,q}_{r,t} \;+\; a^{h,q}\,\tilde{y}_{r,t} \;+\; g^{h,q}\,v^{(7)}_{r,t}\,\left(\tfrac{h}{7}\right)^{\psi^q}, \tag{3}$$

where $\tilde{y}_{r,t}$ is the latest z-scored observation, $v^{(7)}_{r,t} = \tilde{y}_{r,t} - \tilde{y}_{r,t-7}$ is its weekly slope, and $a^{h,q}, g^{h,q}, \psi^q \in (0, 1)$ are learned. The damping exponent ψ^q limits long-horizon slope overshoot (initialised at 0.88 for q=0.5 and 0.5 otherwise). We then invert standardisation.

2.4 Multi-Quantile Pinball Loss

The forecaster minimises the multi-quantile pinball loss [9],

$$\mathcal{L}_{\text{pinball}} = \frac{1}{|\mathcal{Q}|}\sum_{q\in\mathcal{Q}} \mathbb{E}_{(t,h)}\big[\max\big(q\,e,\;(q{-}1)\,e\big)\big], \quad e = \tilde{y}_{r,t+h-1} - \hat{\zeta}^{h,q}_{r,t}, \tag{4}$$

evaluated at all $(h, q) \in \mathcal{H} \times \mathcal{Q}$. The $q = 0.5$ pinball collapses to L1 (point accuracy). Splitting the kernel into its two arms, $\max(qe, (q{-}1)e) = q \max(e, 0) + (1{-}q) \max(-e, 0)$, exposes it as a convex newsvendor loss. The q=0.9 branch penalises under-prediction nine-to-one and is therefore cost-relevant to shortage; the q=0.1 branch has the equal and opposite asymmetry. Because all three branches receive equal weight, their aggregate training objective is not directionally asymmetric. We therefore claim a cost-relevant upper quantile, not an overall cost-asymmetric loss. Training is not through the allocation solver [5]; $\hat{q}^{(0.5)}$ is the raw point forecast and $[\hat{q}^{(0.1)}, \hat{q}^{(0.9)}]$ the raw 80% interval.

We train for up to 800 AdamW epochs (lr $= 10^{-3}$, weight decay 10^{-4}, cosine annealing) on one-day-stride Alpha-wave lookback windows (270 per region, 1,890 in total), with early stopping (patience 80) on the 1,134 Delta-wave validation windows' pinball loss. Once selected, the model is frozen and applied to the Omicron test period via Δt = 7-day rolling origins, yielding 32 origin dates.

2.5 Per-Origin Recalibration

Applied frozen across a regime shift (high-demand Alpha/Delta training, mild Omicron test), the point forecast inherits a systematic over-prediction. At each origin t and horizon h we subtract the mean of the last K=3 realised h-step errors, then blend with seasonal-naive(7) under a per-horizon weight $\alpha \in \{0, 0.1, \ldots, 1\}$ fitted on an expanding window of realised origins (pure model, α=1, until seven origin–region pairs are realised). Both steps use only information available at t.

Let $\hat{\mathbf{q}}^h_{r,t} = (\hat{q}^{(0.1)}, \hat{q}^{(0.5)}, \hat{q}^{(0.9)})$ denote the raw heads and $\tilde{y}^{h,\mathrm{pt}}_{r,t}$ the recalibrated point forecast. Table 1 evaluates both point estimates; the calibration table and every LP instead use $\hat{\mathbf{q}}$ unchanged (apart from clipping negative demand to zero). Thus recalibrated RMSE gains do not enter the allocation results.

3 Regional Allocation Model

The allocation module decides how many surge beds to place in each NHS region and how to route excess demand through inter-regional patient transfers, given the quantile forecasts of Section 2. We present a deterministic linear program, a risk-averse (tail-weighted) extension, and closed-form heuristic baselines.

3.1 Sets, Parameters, and Decision Variables

The seven regions are both demand nodes and capacity sites, $\mathcal{R} = \{r_1, \ldots, r_7\}$. Each already operates critical care, so unlike facility-location models [3] there is no opening decision; the model chooses bed counts and routing. Indices are $r, r' \in \mathcal{R}$; horizons $h \in \mathcal{H} = \{7, 14, 21, 28\}$ days; scenarios $s \in \mathcal{S} = \{\mathrm{low}, \mathrm{median}, \mathrm{high}\}$ from $q^{(0.1)}, q^{(0.5)}, q^{(0.9)}$ with weights $\pi = (0.2, 0.6, 0.2)$, representative design weights for lower, central, and upper stress, not calibrated probability bins. Parameters are demand $\xi^s_{r,h} = \hat{q}^{(s)}_r(t_0 + h)$; baseline critical-care capacity $\Gamma_r = 1.05\times$Delta-peak; surge cap $K_r = 0.5\,\Gamma_r$; budget $B = 0.20 \sum_r \Gamma_r$; great-circle inter-centroid distance $\ell_{rr'}$ in km ($\times 1.3$ road-detour); travel time $T_{rr'} = 60\,\ell_{rr'}/80$ min; mutual-aid cap $\tau = 240$ min; and outbound cap κ (headline case uncapped). The first-stage surge $b_r \in [0, K_r]$, shared across scenarios and horizons. The second-stage flow is $z^s_{rr',h} \geq 0$ (defined for $T_{rr'} \leq \tau$); its self-loop serves demand locally and off-diagonal entries transfer it, with unmet demand $u^s_{r,h} \geq 0$ and a tail auxiliary $W \geq 0$ for the risk-averse variant. The cap admits 12 undirected mutual-aid links.

3.2 Deterministic Median-Scenario LP

For the median scenario m ($q^{(0.5)}$), the deterministic model solves

$$\min \; \textstyle\sum_{r,h} u^m_{r,h} \; + \; \theta \sum_{r \neq r',h} \ell_{rr'}\, z^m_{rr',h}, \tag{5}$$

where $\theta = 10^{-3}$ makes transfer distance a tie-break: one weekly snapshot of unmet demand outweighs 1000 patient-km. The model is subject to

$$\begin{aligned}
z^m_{rr,h} + \textstyle\sum_{r' \neq r:\, T_{rr'} \leq \tau} z^m_{rr',h} + u^m_{r,h} &= \xi^m_{r,h} && \forall r, h, && (6)\\
z^m_{rr,h} + \textstyle\sum_{r' \neq r:\, T_{r'r} \leq \tau} z^m_{r'r,h} &\leq \Gamma_r + b_r && \forall r, h, && (7)\\
\textstyle\sum_{r' \neq r} z^m_{rr',h} &\leq \kappa && \forall r, h, && (8)\\
\textstyle\sum_r b_r &\leq B, && && (9)\\
z^m_{rr',h} &= 0 && \text{if } T_{rr'} > \tau, && (10)\\
b_r, z^m_{rr',h}, u^m_{r,h} &\geq 0. && && (11)
\end{aligned}$$

Equation (6) partitions demand into local service, reachable transfers, and unmet demand. Equations (7) and (8) enforce bed and outbound-service limits; Equations (9) and (10) enforce surge and mutual-aid limits. Minimisation leaves $u^m_{r,h}$ at the residual shortfall. These are weekly snapshots summed across four weeks, not a bed-day integral.

3.3 Risk-Averse Tail-Weighted Objective

To hedge against the high-demand scenario the risk-averse model imposes Equations (6) to (8) for every $s \in \mathcal{S}$, retains Equations (9) and (10), and adds a tail term $\lambda_3 W$ ($W \geq \sum_{r,h} u^s_{r,h}$ for $s \in \mathcal{S}_{\text{tail}} = \{\text{high}\}$), giving the objective

$$\min \sum_{s\in\mathcal{S}} \pi_s\Big(\textstyle\sum_{r,h} u^s_{r,h} + \theta \sum_{r\neq r',h} \ell_{rr'} z^s_{rr',h}\Big) \; + \; \lambda_3 W. \tag{12}$$

This scenario-weighted LP up-weights high demand; it is not robust optimisation formally. With one tail scenario, $W = \sum_{r,h} u^{\text{high}}_{r,h}$, so $\lambda_3 W$ is an extra high-tail penalty, not conditional value-at-risk. We call it the robust LP for brevity and report λ_3=1 (λ_3=0 recovers the risk-neutral objective and $\lambda_3 \to \infty$ a worst-case policy on the high scenario); the tail term binds only under tight budgets (Section 5.3). Imposing $q^{(0.9)}$ jointly across regions and horizons makes the construction co-monotone and over-conservative relative to a joint distribution. We denote the unmet component by $U_\pi = \sum_s \pi_s \sum_{r,h} u^s_{r,h}$ and call it *scenario-weighted unmet demand*. Because π contains design weights rather than estimated joint probabilities, U_π is not an estimate of real-world expected shortage.

3.4 Exact LP Solution and Baselines

Both models are continuous LPs, solved exactly in PuLP/CBC in <0.1 s; deployed surge counts can be rounded to integers. Applying metaheuristics to a larger trust-level formulation is left for future work (Section 6). For every fixed surge $\mathbf{b}^\star$, a common residual LP fills in (z, u) across all scenarios, making U_π and u^{worst} comparable across policies.

Finally, three closed-form heuristics give a no-optimisation floor. Population-proportional uses ONS region populations, demand-proportional uses scenario-weighted peak demand, and greedy shortage-first adds single-bed increments to the most-unmet region. Each fixes $\mathbf{b}^\star$ and is then completed by the same LP step.

4 NHS England Case Study

The case study uses two public sources covering 1 Aug 2020 – 31 Aug 2022 (the period of daily NHS regional publication). These are NHS England COVID-19 Hospital Activity [11], which provides daily hospital and mechanical-ventilation bed occupancy at NHS regional resolution; and ONS Mid-Year Population

Estimates [12] on 2021 geography, used as the per-region PINN normaliser and the population-proportional baseline denominator. All experiments use the seven NHS England commissioning regions. The series is split chronologically by epidemic wave so the most clinically distinct wave is held out, with training on 1 Aug 2020 – 31 May 2021 (Alpha and early vaccination), validation on 1 Jun – 30 Nov 2021 (Delta) for hyper-parameter selection and early stopping, and testing on 1 Dec 2021 – 31 Aug 2022 (Omicron and post-Omicron). Omicron's case-to-hospitalisation and hospitalisation-to-critical-care ratios differ markedly from those of earlier waves, providing a regime-shift stress test.

For the allocation problem, baseline capacity Γ_r is *measured* as each region's Delta-wave peak MV-bed occupancy $\times 1.05$, whereas the per-region surge cap $K_r = 0.5\,\Gamma_r$, total budget $B = 0.20 \sum_r \Gamma_r \approx 213$ beds (the surge headroom above the $\sum_r \Gamma_r \approx 1065$-bed measured baseline, not existing beds), mutual-aid travel cap $\tau = 240$ min, and uncapped outbound transfers ($\kappa = \infty$) are planning assumptions. The surge b_r is the model's decision, the extra MV beds a region stands up above its measured baseline. The forecaster, budget B, tail weight λ_3, travel cap τ, and transfer cap κ are each swept in Section 5.3. Forecast metrics are macro-averaged across the seven regions and the 32 rolling-origin evaluations spanning the Omicron test period; the full solver comparison is reported at the first Omicron decision origin (29 Dec 2021), with an all-origin check in Section 5.3. These are stress-test settings, not NHS standards: the 5% uplift is a headroom proxy, K_r prevents unlimited concentration, and the 20% budget and 240-min window form a generous headline case. Deployment requires staffed-bed inventories, local expansion and transfer limits, network travel times, and joint residuals for scenario probabilities; our proxy supports policy comparison, not deployable bed-count claims.

5 Results and Discussion

5.1 Forecasting Accuracy

Table 1 uses 32 rolling origins; MASE divides MAE by in-sample seasonal-naive(7) MAE. Raw PinnGRU leads at h=7, 14, with a paired Wilcoxon gain over ARIMA at h=14 (-3.88 beds, p=0.003). Regime-shift over-prediction then hurts: at h=28 its underestimation rate is 10.3% and it trails ARIMA by 5.53 beds (p=0.006). The model-agnostic recalibration leads the unrecalibrated baselines at h=7–21 and nearly ties ARIMA at h=28 (28.5 vs 28.0), with underestimation rising to 37%. Its gains are point-only; the raw LP stress scenarios are evaluated below. Lag-only XGBoost and GRU degrade most under Omicron's altered severity, whereas per-origin ARIMA refitting is steadier.

Three ablations against the raw PinnGRU (h=14 RMSE 14.46, MASE 0.77) isolate the load-bearing parts. Replacing the pinball loss with a mean-squared-error (MSE) loss lifts h=14 RMSE to 18.70, skipping PINN pre-training to 17.27, and zeroing the four PINN parameter features to 27.25, the largest regression. The structured PINN-derived representation adds signal, but its non-identifiable parameters cannot separate physical structure from representation capacity.

Table 1. Forecast accuracy (best **bold**, second best <u>underlined</u> per column).

Model	RMSE by horizon (days)				MASE
	7	14	21	28	(all h)
Seasonal-naive(7)	13.53	18.16	<u>22.31</u>	**25.98**	0.63
ARIMA per region	11.33	17.80	23.41	<u>28.02</u>	<u>0.61</u>
XGBoost (lag features)	12.45	22.90	36.42	58.88	0.89
GRU per region	14.13	22.92	33.35	43.97	0.90
PinnGRU (raw)	<u>11.02</u>	<u>14.46</u>	24.19	34.63	0.77
PinnGRU+recal (ours)	**10.56**	**13.34**	**20.79**	28.54	**0.56**

Table 2. Raw PinnGRU quantile evaluation by horizon (224 region–origin forecasts each). Coverage and miss are percentages; width and scores are in MV-bed units.

h	Cov. 80	Width	P.1	P.5	P.9	WIS	q.9 miss	Cross
7	93.8	44.3	1.75	4.52	2.95	6.15	0.4	0.0
14	83.9	55.1	1.84	6.02	4.44	8.20	0.9	0.0
21	77.2	70.8	2.72	10.11	6.08	12.61	0.4	0.0
28	63.8	89.8	4.84	14.36	8.36	18.38	0.0	0.0

Table 2 scores the unrecalibrated LP heads. Coverage falls from a conservative 94% at 7 days to 64% at 28 as $q^{(0.1)}$ exceeds truth increasingly $(5.8\%, 15.2\%, 22.3\%, 36.2\%)$. The $q^{(0.9)}$ miss rate is only 0–0.9%, so "high" is a stress level, not a calibrated event. The independent heads are not order-constrained, but none of 896 forecasts crosses; no rearrangement is applied and the LP only clips negative demand to zero.

5.2 Allocation Performance Across Solvers

Table 3 compares the allocation policies on the first Omicron origin with $B = 213$ beds and a four-hour cap. The common three-scenario residual LP makes U_π and u^{worst} comparable. With $\kappa{=}\infty$, all full-budget policies tie on coverage; the robust LP's 672 patient-km is 4.6× below population-proportional and 1.7× below greedy. The deterministic LP uses only 51 beds and incurs about 3× greater shortage, showing the value of scenario aggregation. Capping transfers at $\kappa{=}10$ breaks the coverage tie. The per-region structure in Figure 3a shows the deterministic LP placing 50.9 of its 51.4 beds in South East and 0.4 in East of England; the robust LP uses its full budget across six regions.

5.3 Forecast-Quality and Parameter Sensitivity

Substituting each forecaster into the robust LP at the 29 Dec 2021 origin, every $q^{(0.9)}$ peak sum except seasonal-naive exceeds the 797-bed realised peak. Realised unmet is zero, so forecast quality buys efficiency rather than coverage in this mild instance. Across all 32 Omicron origins, robust, population-proportional, and demand-proportional plans share mean $U_\pi = 2.9$, but the robust LP cuts transfer burden to 76 patient-km versus 406 and 337. The proportional rules always fill

Table 3. Allocation comparison under $B = 213$ beds, $\tau = 240$ min, and three PinnGRU scenarios; the κ=10 column re-scores each policy under a capped transfer service.

Policy	**Beds**	κ=∞		κ=10	**Transfer**	**Time**
		U_π	u^{worst}	U_π	**(patient km)**	**(s)**
Population-proportional	212.7	29.0	144.8	48.8	3119.8	0.00
Demand-proportional	212.7	29.0	144.8	51.4	2615.3	0.00
Greedy shortage-first	212.7	29.0	144.8	**32.3**	1140.0	9.98
Deterministic LP	51.4	83.3	416.4	84.4	1175.1	0.03
Robust LP (λ_3=1)	212.7	29.0	144.8	**32.3**	**671.7**	0.04

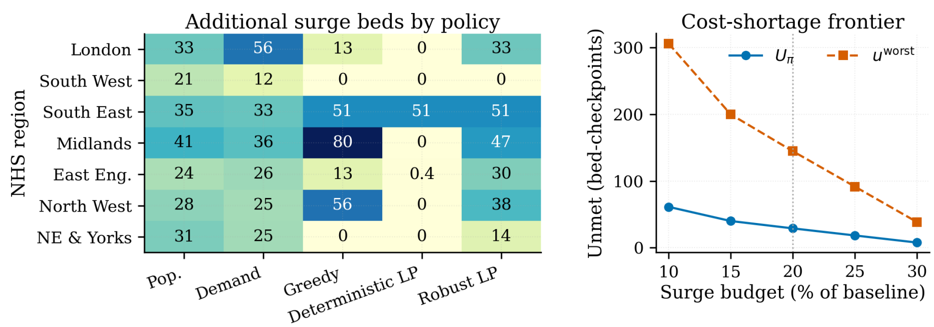

Fig. 3. Allocation structure and robust-LP budget frontier. (a) Additional surge beds allocated by each policy at the first Omicron origin (0 denotes no added beds in that region). (b) Scenario-weighted unmet U_π and worst-case unmet u^{worst}, summed across four forecast checkpoints, versus the surge budget.

212.7 beds, whereas the robust LP averages 35.4, so the routing-efficiency gain holds across origins.

In one-at-a-time sweeps, budget B dominates (Figure 3b). Cutting $B/\sum_r \Gamma_r$ from 0.20 to 0.10 doubles U_π and triples u^{worst}, while 0.30 cuts U_π to $\sim 1/4$. Only the high scenario has residual shortfall throughout the sweep. Consequently, $U_\pi = \pi_{\text{high}} u^{\text{worst}} = 0.2 u^{\text{worst}}$, so the curves show the same trend at different scales. λ_3 is *non-binding* at $B = 0.20$: the weighted and high-tail objectives share an optimizer; only a tighter budget ($\leq 15\%$) re-engages it. The transfer controls τ and κ determine when placement matters. With uncapped recourse on 12 links, coverage depends only on total surge and optimisation buys routing. Pruning to five links (τ=120 min) puts the robust LP ahead of every heuristic on scenario-weighted and worst-case unmet (29.9 vs 32.3–34.9 at B=20%), and capping the transfer service at κ=10 gives the same separation (the κ=10 column of Table 3), where greedy matches the LP only by issuing 213 oracle calls at 1.7× its transfer burden. Thus placement matters when transfers are scarce.

6 Conclusion

We presented a forecast-to-decision pipeline for NHS critical-care surge planning that couples a per-region physics-informed forecaster to a risk-averse LP.

PinnGRU leads at two weeks (19% over ARIMA, 37% over GRU), and point recalibration extends competitiveness. Yet raw quantile scoring reveals a conservative high scenario and deteriorating long-horizon coverage, so allocation outcomes are stress tests, not real-world expectations. Under scarce transfers the robust LP is never beaten on coverage.

Limitations are one NHS system, occupancy-based capacity, illustrative parameters, detour-scaled travel, and co-monotone tails. Future work will recalibrate all heads, estimate correlated scenarios, compare upper-weighted and decision-integrated losses with equal weighting, and use a matched-capacity time-encoder control. Other health systems and a trust-level instance large enough for metaheuristics are also needed.

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